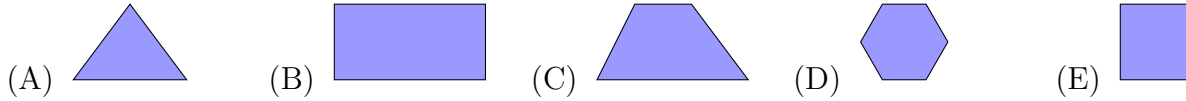


GRADES 5 AND 6

SAMPLE QUESTION FOR 3 POINTS

Which of the following shapes cannot be divided into two triangles by a single straight line?



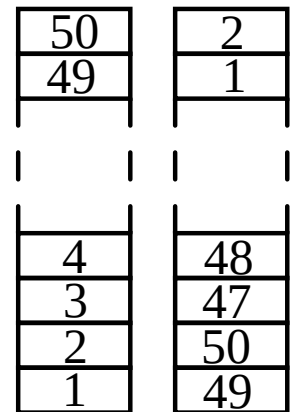
SAMPLE QUESTION FOR 4 POINTS

Maria, Peter, Richard, and Tina were playing football in the classroom and one of them broke a window. When the principal asked who did it, she got the following responses: Maria: “It was Peter.” Peter: “It was Richard.” Richard: “It wasn’t me.” Tina: “It wasn’t me.” Only one child was telling the truth. Who broke the window?

- (A) Maria (B) Tina (C) Peter (D) Richard
(E) It can’t be determined with certainty.

SAMPLE QUESTION FOR 5 POINTS

On a table there is a tower made of blocks numbered from 1 to 50. Emma builds a new tower in the following way. She takes two blocks from the top of the original tower and puts them on the table as the base of the new tower. She continues by taking the two top blocks from the remainder of the original tower and putting them on the top of the new tower, as seen in the diagram. Which of the following pairs of numbers are on adjacent blocks in the new tower?

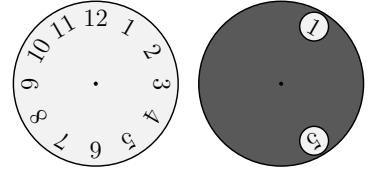


- (A) 29 and 28 (B) 34 and 35 (C) 29 and 26
(D) 31 and 33 (E) 27 and 30

GRADES 7 AND 8

SAMPLE QUESTION FOR 3 POINTS

A gray circle with two holes in it is placed on top of a clock face, as shown. The gray circle is turned around its center so that an 8 appears in one hole. Which two numbers could be seen in the other hole?



- (A) 4 or 12 (B) 1 or 5 (C) 1 or 4 (D) 7 or 11 (E) 5 or 12

SAMPLE QUESTION FOR 4 POINTS

Anika wrote down three consecutive whole numbers in increasing order, but instead of digits she used symbols so wrote $\square\diamond\diamond$, $\heartsuit\triangle\triangle$, $\heartsuit\triangle\square$. What would she write next?

- (A) $\heartsuit\heartsuit\diamond$ (B) $\square\heartsuit\square$ (C) $\heartsuit\triangle\diamond$ (D) $\heartsuit\diamond\square$ (E) $\heartsuit\triangle\heartsuit$

SAMPLE QUESTION FOR 5 POINTS

Bart wrote the number 1015 as a sum of numbers using only the digit 7. He used a 7 a total of 10 times, as shown. Now he wants to write the number 2023 as a sum of numbers using only the digit 7, using a 7 a total of 19 times. How many times will he use the number 77?

$$\begin{array}{r} 777 \\ 77 \\ + 77 \\ 77 \\ 7 \\ \hline 1015 \end{array}$$

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6